4.4 Distance on the Coordinate Plane

Lesson Introduction

Benchmark	MA.6.GR.1.2 Find the distances between ordered pairs, limited to the same x-coordinate or the same y-coordinate, represented on the coordinate plane.	Learning Target	• I can find the distance between two ordered pairs.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.8.GR.1.2 Apply the Pythagorean theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Essential Questions	• How can you find the distance between two ordered pairs whose y-coordinates are the same? • How can you find the distance between two ordered pairs whose x-coordinates are the same? • How is absolute value related to finding the distance between two ordered pairs?
Lesson Narrative	In Lesson 4, students will begin developing their understanding of finding distance on the coordinate plane. Students will build upon knowledge of graphing opposites on both vertical and horizontal number lines. You will lead students toward understanding the relationship between finding the distance between values on the coordinate plane to operations with absolute value.

Component Summary	4.4.1 Warm-Up (~5 min.)	Plot rational numbers on a vertical number line *(SE p. 151)*	4.4.4 Guided Practice (10 min.)	Find distance between two points using addition and subtraction of coordinates *(SE p. 155)*
	4.4.2 Collaboration (5 min.)	Collaborate with a partner on how to find distance between two points *(SE p. 152)*	4.4.5 Your Turn! (10-15 min.)	Find the distance between two points *(SE p. 156)*
	4.4.3 Guided Instruction (15-20 min.)	Count on a grid to find distance between two points *(SE p. 153)*	4.4.6 Wrap-Up (~5 min.)	Create a summary of how to find distance between two points *(SE p. 158)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Absolute value Additional Terms: • Quadrant	• (Optional) Graph paper	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may confuse when to add the absolute values and when to subtract the absolute values when finding the distance between two points. For example, when finding the difference between two points, if one is negative, students may believe that negative represents the difference, then add the two values.</td><td>Students have not been exposed to operations with integers prior to this unit. Therefore, students cannot use those operations to find the distance between points with negative coordinates. A foundation for operations with integers can be laid by verbally expressing the subtraction while writing the expression. If students are naturally understanding the operations, then the teacher can further explain. But student understanding of the operations is not necessary for their success in this lesson.
Consider bringing the students back to the visual of the coordinate plane when determining whether to add or subtract to find the distance.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Think-Pair-Share
Consider implementing a Think-Pair-Share for 4.4.2 Collaboration. This collaboration positions students to think about how they would find distance between two coordinates and share their thoughts with a partner. Consider supplying a sentence starter such as “I think _____ is correct because…”.</td><td>MA.K12.MTR.6.1: Reasonableness
When finding distance between two points, students should use reasoning and sensemaking to determine whether their answer is correct or not. Consider encouraging students to check every answer for reasonableness in order to verify their answers. Ask questions like “Does this answer make sense in this context?”.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Guess My Point
Consider providing students with an alternative way to continue applying their knowledge of the coordinate plane and finding distance with a game of Guess My Point. Similar to Guess Who?, students in pairs must ask one yes or no question at a time to determine the secret location of their partner’s ordered pair. Once both students have discovered the secret location, they must work together to find the distance between their locations. This can be done by mapping your classroom as well, with students picking secret objects around the room that have been designated an ordered pair as a location. Read over 4.4.5 Your Turn! to see an example.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

4.4.1 Warm-Up: A Point on the Number Line

Component Overview: *Students plot rational numbers on a vertical number line.*

4.4.1 Warm-Up: A Point on the Number Line

1. Circle all numbers that could be the location of point F on the number line.

3.5	$4\frac{1}{2}$
$-\frac{3}{5}$	-3.5
$-\frac{7}{2}$	$-4\frac{1}{2}$

Connect the vertical number line to the coordinate plane that students have been working with throughout this unit and will continue to work with in this lesson.

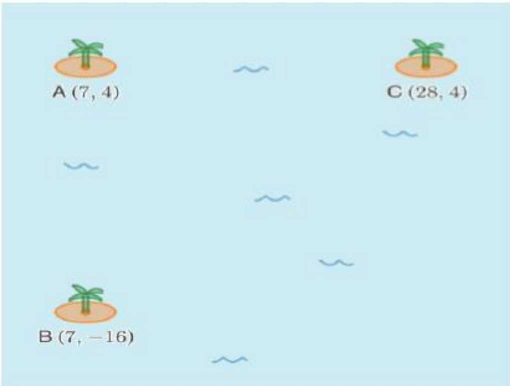
- *How is the vertical number line similar to the coordinate plane?*
- *How can you use what you know about number lines to plot points on the coordinate plane?*

4.4.2 Collaboration: Who Is Correct?

Component Overview: *Students will collaborate with a partner to determine the distance between three islands. They are given coordinates of each island and must use those and their mathematical reasoning to determine which student is correct about the islands' distances from one another.*

4.4.2 Collaboration: Who Is Correct?

The locations of three islands in the ocean are given using ordered pairs.



Three students described the distance between each island.

Braxton says the distance from point A to point C is **less than** the distance from point A to point B.

Justine disagrees with him and says the distance from point A to point C is **greater than** the distance from point A to point B.

Rainey says the distance from point A to point C is **equal to** the distance from point A to point B.

1. Discuss with your partner who you think is correct. Write a summary of your discussion.
   Answers will vary. Justine is correct because Island A is only 20 units away from Island B, while it is 21 units away from Island C.

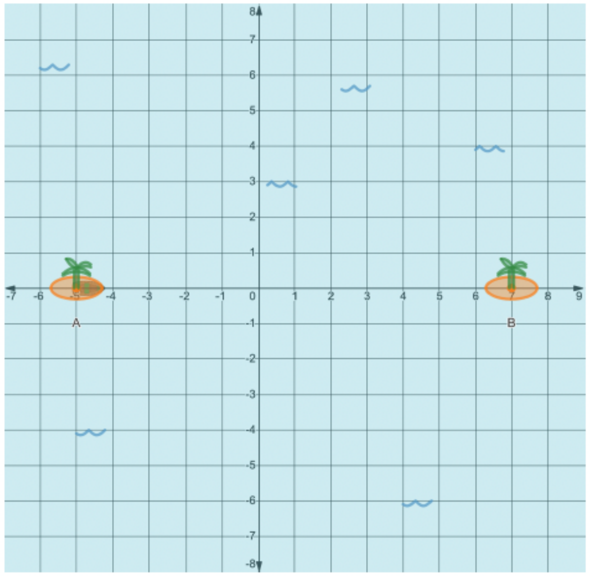
Think-Pair-Share
Consider implementing a Think-Pair-Share for 4.4.2 Collaboration. This Collaboration positions students to think about how they would find the distance between two coordinates and share their thoughts with a partner.
"I think ____ is correct because…"
Spend time probing students on their reasoning for which student they think is correct. Highlight student answers that use the coordinates and add or subtract to find distance.

4.4.3 Guided Instruction: Distance on the Coordinate Plane

Component Overview: *Students use counting grid lines to find the distance between two islands or two points plotted on the coordinate plane. Students must also be able to identify the coordinates of the points and use those coordinates to discover how to find distances between points on the coordinate plane.*

4.4.3 Guided Instruction: Distance on the Coordinate Plane

1. The coordinates of two small islands are labeled points A and B on the graph.



   a. Counting the grid lines, how far apart are the islands?
   12 *units*

   b. What are the coordinates of points A and B?
   A: $(-5,0)$
   B: $(7,0)$

   c. Explain why the y-coordinates of the two points are the same.
   Answers will vary. The y-coordinates are the same because the islands are horizontal to one another, and both islands are on the x-axis.

Notice and Wonder
Engage students in a Notice and Wonder routine to review the vocabulary and components of the coordinate grid.

- *What do you notice?*
- *What do you wonder about the two islands?*
- *What axis are the islands on? Label it. Label the other axis.*
- *Are the islands the same distance from the origin?*

Some students may confuse how to write the coordinates correctly. If students write $(0, -5)$, review where the x- and y-coordinates go in an ordered pair (x, y).

d. Describe the location of the two points relative to the y-axis.
Island A is 5 spaces to the left of the y-axis
Island B is 7 spaces to the right of the y-axis

The distance a number is from zero (0) on a number line is called the ***absolute value***.

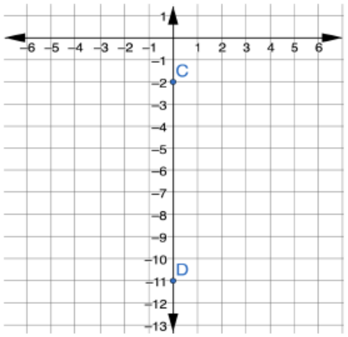
e. What is the absolute value of the x-coordinate of each point?

The absolute value of the x-coordinate of Point A is _5_.

The absolute value of the x-coordinate of Point B is _7_.

f. Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.
$|-5|+|7|=12$

2. Below is another coordinate plane with two different islands located at points C and D.



a. How is the location of these two islands different from the first two you worked with?
These islands are both on the y-axis and the other two were both on the x-axis.

b. Counting the grid lines, how far apart are the islands?
9 units

c. What are the coordinates of points C and D?
C: $(0, -2)$
D: $(0, -11)$

Solidify student understanding through questions like: Do you think this will always be true for two points on a horizontal line? Why or why not?

Students may need a reminder on how to write a mathematical sentence using absolute value.

Consider prompting by writing the following on the board:

|____| + |____| = __________

4.4.3 Guided Instruction: Distance on the Coordinate Plane (continued)

d. Explain why the x-coordinates of the two points are the same.
Answers will vary. The x-coordinates are the same because the two points are on the same vertical line. The two points are both on the y-axis.

e. Describe the location of the two points relative to the x-axis.
Point C is 2 spaces below the x-axis
Point D is 11 spaces below the x-axis

f. What is the absolute value of the y-coordinate of each point?
The absolute value of Point C is _2_.
The absolute value of Point D is _11_.

g. Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.

$$|-11| - |-2| = 9$$

3. Complete the statements.
When finding the horizontal distance between two points that lie on the same line, if the points are on opposite sides of the y-axis, (○ add / ○ subtract) the absolute value of the. unlike coordinates. If the points are on the same side of the y-axis, then (○ add / ○ subtract) the absolute values of the unlike coordinates.
When finding the vertical distance between two points that lie on the same line, if the points are on the opposite sides of the x-axis, (○ add / ○ subtract) the absolute value of the unlike coordinates. If the points are on the same side of the x-axis, then (○ add / ○ subtract) the absolute value of the unlike coordinates.

Consider creating a 3D number line on the floor and having students walk the number line to count the distance between two points. Ask the class to count the distance out loud. Have students try a few examples where they are on the same side of the number line and a few where they are on different sides of the number line.

Solidify student understanding through intentional questions like:
Do you think this will always be true for two points on a vertical line? Why or why not?

Notice and Wonder
Have students illustrate/draw/sketch as they read the paragraph. Once they have sketched what they read, prompt students to notice the similarities and differences based on location in quadrants.

- *What do you notice?*
- *What is the same? What is different?*
- *What does that make you wonder or think about when finding distance on the coordinate plane?*

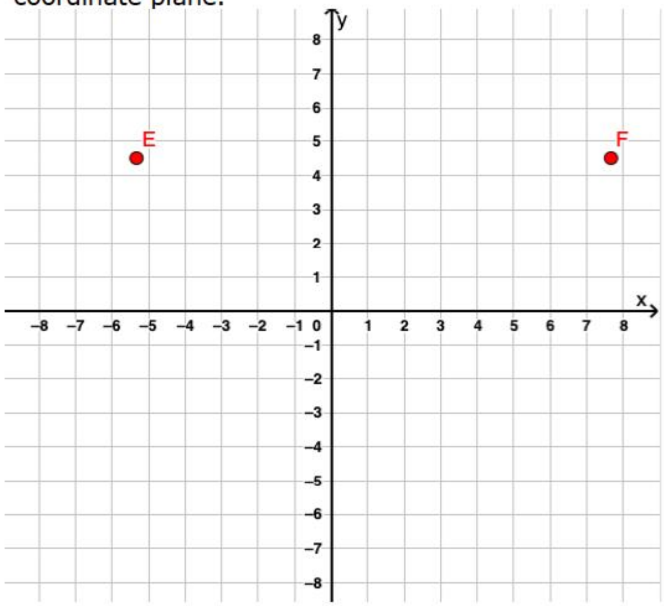
Monitor for student responses that include:
- *"A pattern I noticed among the coordinates of the points in the same quadrant is…"*
- *Adding or subtracting the coordinates to find distance.*

4.4.4 Guided Practice: Distances Between Rational Numbers

Component Overview: *Students practice finding distance between two points on the coordinate plane using addition or subtraction.*

4.4.4 Guided Practice: Distances Between Rational Numbers

1. Graph the points $E\left(-5\frac{1}{3}, 4\frac{1}{2}\right)$ and $F\left(7\frac{2}{3}, 4\frac{1}{2}\right)$ on the coordinate plane.



a. Describe the location of these points relative to the origin and the quadrants they lie in.
Answers will vary. Point E is $5\frac{1}{3}$ units left and $4\frac{1}{2}$ units up from the origin in quadrant II.
Point F is $7\frac{2}{3}$ units right and $4\frac{1}{2}$ units up from the origin in quadrant I.

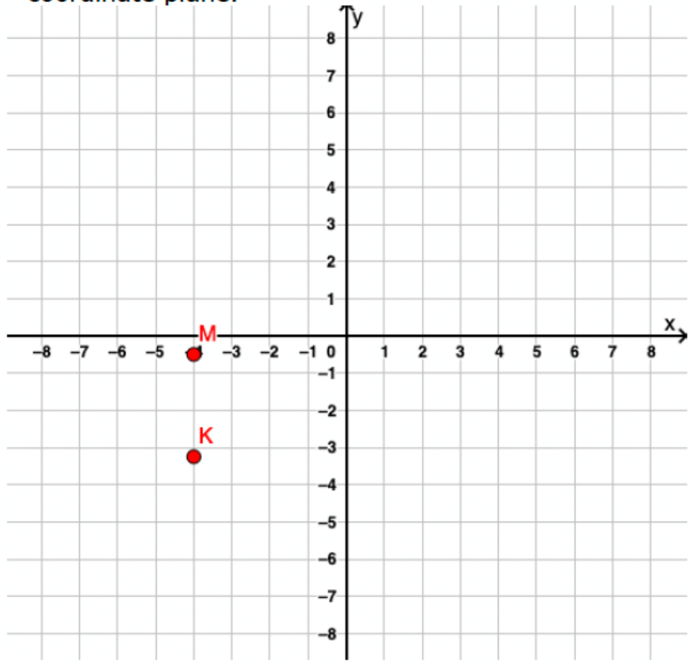
b. What is the distance between point E and point F?
$\left|-5\frac{1}{3}\right| + \left|7\frac{2}{3}\right| = 12\frac{3}{3}$ *or* 13

Florida's assessment limits for MA.6.GR.1.2 and MA.6.GR.1.3 state that all ordered pairs should be integers. The rational ordered pairs in 4.4.4 and 4.4.5 help to support the area of emphasis by requiring students to perform all four operations with integers, positive decimals and positive fractions with procedural fluency, while also continuing student understanding of finding distance. By stretching the student's experience beyond integers, the student will not necessarily be able to rely on counting to determine the distance, thus solidifying their understanding of absolute value as the distance from zero.

Check for understanding of plotting fractions on the coordinate plane. Misconceptions might include students plotting $-5\frac{1}{3}$ at $\frac{1}{3}$ units to the right of -5 instead of to its left.

4.4.4 Guided Practice: Distances Between Rational Numbers (continued)

2. Graph the points $K\ (-4, -3.25)$ and $M\ (-4, -0.5)$ on the coordinate plane.



   a. Describe the location of these points relative to the origin and the quadrants they lie in.
   Answers will vary.
   Point K is 4 units left and 3.25 units down from the origin in quadrant 3.
   Point M is 4 units left and 0.5 units down from the origin in quadrant 3.

   b. What is the distance between point K and point M?
   $|-3.25| - |-0.5| = 2.75$

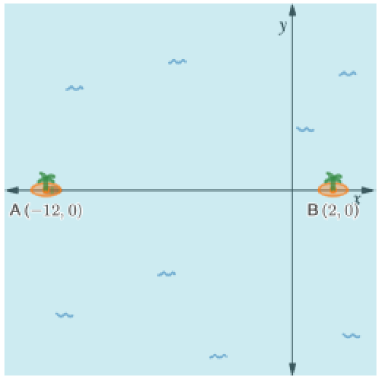
Consider giving students only the coordinates and having them find the distance without the visual of the coordinate plane.

4.4.5 Your Turn!

Component Overview: *Students practice finding distance between two points on the coordinate plane using addition and subtraction.*

4.4.5 Your Turn!

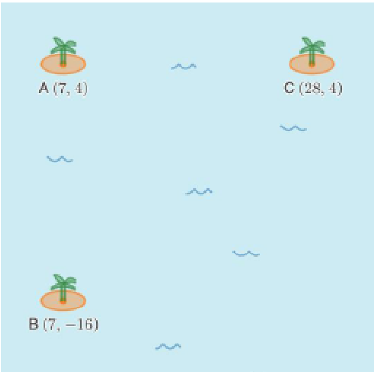
1. What is the distance from point A to point B?



Point A is 14 units away from point B. $|-12| + |2| = 14$

Notice that in question 1 students don't have the grid lines to count the spaces between the two islands. This will prove difficult for some students who might have been using this method. Prompt those students to determine how they might find the distance without counting. Once they have answered, encourage them to then use the counting method to verify their answer.

2. Explain which island is closest to island A. Show your work.



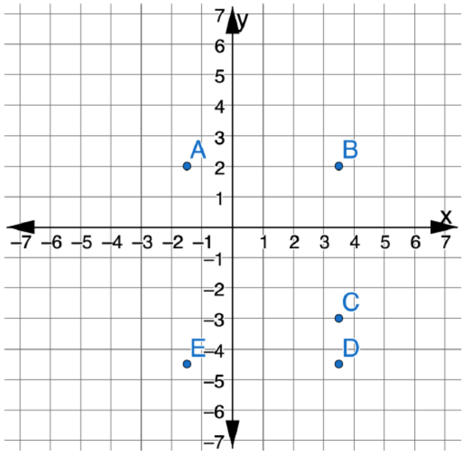
Answers will vary.
Distance between point A and B: $|-16| + |4| = 20$
Distance between point A and C: $|28| - |7| = 21$
Island B is closest to island A because they are only 20 spaces apart, while island C is 21 spaces from island A.

Check for students using mathematical reasoning to explain their answers. Consider providing sentence frames for students to explain their reasoning:

- *Island ___ is closest to Island A because _____.*
- *I used adding/subtracting to show that Island ___ is closest to Island ___ by ____ units.*

4.4.5 Your Turn! (continued)

3. Five points are shown. Points A and E lie on the same vertical line. Points B, C, and D lie on a different vertical line. Points E and D lie on the same horizontal line.



a. Complete the table by filling in each missing coordinate.

A	$(-1.5, 2)$
B	$(3.5, 2)$
C	$(3.5, -3)$
D	$(3.5, -4.5)$
E	$(-1.5, -4.5)$

b. The distance between points B and C is 5 units.

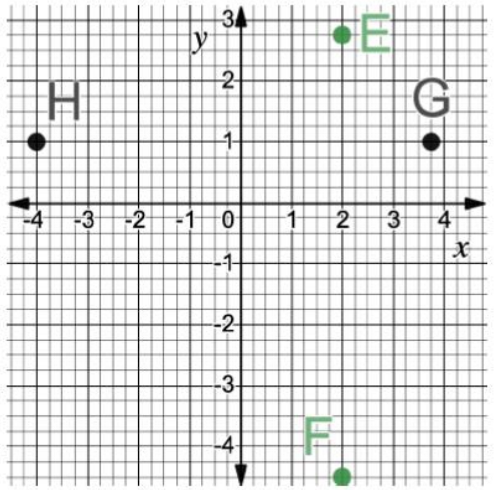
c. The distance between points D and C is 1.5 units.

d. The distance between points D and E is 5 units.

Guess My Point
For students who finish early or for an alternative method for engaging students in the practice of plotting and finding distance, consider having students play the Guess My Point activity.

- *Each partner picks a secret location for their point. (Choose an ordered pair.)*
- *Students then take turns asking one question at a time to determine the location of their partner's point (similar to Guess Who?). Encourage students to ask questions that incorporate vocabulary from the lesson. For example, "Is your point in Quadrant 1?" or "Does your location have a positive x-coordinate?"*
- *Once both students have discovered the location of their partner's point, they must collaboratively find the distance between those two points.*

4. Use the coordinate plane below to determine distances between the following points.



a. The distance between points E and F is _7.25_ units.

b. The distance between points G and H is _7.75_ units.

c. Which points are the same distance from $(2, 1)$?
Points E and G are the same distance from the point $(2, 1)$.
Point E is 1.75 units from the point $(2, 1)$.
Point G is also 1.75 units from the point $(2, 1)$.

Notice the answer here is in fourths. Consider prompting students to identify the scale first when reading the graph. Students should notice that each grid line is an increase of $\frac{1}{4}$.

4.4.6 Wrap-Up: Cool Down

Component Overview: *Students write a summary for a friend about how to find the distance between two points on the coordinate plane.*

4.4.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them learn how to find the distance between two points on a coordinate plane.
   Answers will vary.

For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson.

- *Recording a message explaining the concept or examples verbally*
- *Drawing and labeling, providing an example*
- *Constructing a graphic organizer*